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Definition and Acronyms

Acronym	Definition
AI	Artificial Intelligence

API	Application Programming Interface
APK	Android Package Kit
BA	Business Analyst
CRUD	Create, Read, Update, Delete
DB	Database
DI	Dependency Injection
ERD	Entity Relationship Diagram
FE	Functional Element
HKD	Hộ Kinh Doanh
IDE	Integrated Development Environment
KPI	Key Performance Indicator
LAN	Local Area Network
LLM	Large Language Model
ORM	Object-Relational Mapping
POS	Point of Sale
PR	Pull Request
RAG	Retrieval-Augmented Generation
RDBMS	Relational Database Management System
SDD	Software Design Description
SEO	Search Engine Optimization
SRS	Software Requirement Specification
SST	Server-Side Rendering
STT	Speech-to-Text
TC	Test Case
UAT	User Acceptance Testing
UI	User Interface
UX	User Experience
WBS	Work Breakdown Structure

I. Project Introduction

1. Overview

1.1. Project Infomation

- **Project Name:** Platform to support digital transformation for household businesses.
- **Vietnamese Name:** Nền tảng hỗ trợ chuyển đổi số cho hộ kinh doanh.
- **Abbreviation:** BizFlow.
- **Group Name:** 6Members.
- **Software Type:** Web application & Mobile application

1.2. Project Team

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2. Product Background

In Vietnam, household businesses play a critical role in the local economy, particularly in traditional sectors such as building materials, construction supplies, and hardware retail. Most of these fall under Group 1 or Group 2 classifications defined by the Ministry of Finance's Decision 3389/QĐ-BTC (2025).

Currently, the majority of these businesses operate using fully manual workflows. Daily tasks such as recording sales, managing inventory, tracking customer debts, and processing orders from phone calls or Zalo are typically performed using handwritten notebooks or simple Excel files. Furthermore, these businesses often lack the budget to hire professional accountants, leading to difficulties in financial management and tax compliance.

3. Existing Systems

Despite the growing demand for digital transformation, existing commercial POS or business management solutions in the market often fail to meet the unique needs of household businesses.

General Drawbacks of Existing Solutions:

- **Not Optimized for Household Scale:** Most systems are designed for restaurants, retail fashion chains, or large enterprises, making them too complex for small store owners.
- **Hardware Dependency:** Existing POS solutions typically require multiple devices (computers, barcode scanners, receipt printers), while many household businesses operate with only a single smartphone.
- **Lack of Multi-channel Support:** They do not effectively handle informal order channels like phone calls or Zalo messages, which are prevalent in traditional trade.
- **Weak Debt Management:** They often lack specialized features for tracking long-term customer transaction history and debts.
- **Low Digital Literacy Compatibility:** The user interfaces are often too complicated for store owners with limited digital skills.

4. Business Opportunity

Due to the lack of a specialized platform, household businesses face frequent errors in manual calculations, slow order processing, inventory discrepancies, and inconsistent debt records.

BizFlow addresses this gap by providing a comprehensive system designed specifically for traditional stores. By integrating an AI-powered assistant capable of understanding natural language (via text or voice), the system allows users to create draft orders automatically. Additionally, the auto-bookkeeping feature ensures compliance with **Circular 88/2021/TT-BTC**, helping business owners manage finances accurately without needing deep accounting knowledge.

5. Software Product Vision

BizFlow aims to be the ultimate digital assistant for household businesses in Vietnam. The vision is to replace manual notebooks with an intelligent, automated, and easy-to-use platform that:

- Reduces human errors in sales and accounting.
- Supports automation through AI voice processing.
- Provides real-time visibility into business operations (revenue, inventory, debt).
- Ensures legal compliance with Vietnamese accounting standards for household businesses.

6. Project Scope & Limitations

6.1. Major Features

6.1.1. Major Features for Employee (Mobile App)

- **FE-01:** Login to the system.
- **FE-02:** Create at-counter orders quickly (search products, select quantity).
- **FE-03:** Record debt for registered customers directly during order creation.
- **FE-04:** Print sales orders using pre-designed bill templates.
- **FE-05:** Receive real-time notifications when AI/Chatbot generates a new order.
- **FE-06:** View and confirm "Draft Orders" created by the AI assistant.

6.1.2. Major Features for Owner (Web Application)

Includes all Employee permissions, plus:

- **FE-07:** Manage Product Catalog (create, update, disable products; define units of measure).
- **FE-08:** Manage Inventory (record stock imports, view stock levels, inventory history).

- **FE-09:** Manage Customers (view info, purchase history, track outstanding debts).
- **FE-10:** View Reports & Analytics (daily/monthly revenue, best-sellers, debt reports).
- **FE-11:** Manage Employee Accounts (create, reset password, deactivate).

6.1.3. Major Features for System & AI

- **FE-12: Voice-to-Order:** Convert natural language (voice/text) into a structured draft order.
- **FE-13: Auto-Bookkeeping:** Automatically perform bookkeeping for sales, imports, and debt transactions.
- **FE-14: Compliance Reporting:** Automatically generate accounting reports following **Circular 88/2021/TT-BTC**.

6.1.4. Major Features for Administrator (Web Portal)

- **FE-15:** Manage Owner accounts (activate/deactivate).
- **FE-16:** Manage Subscription Pricing.
- **FE-17:** View Platform Analytics & Feedback.
- **FE-18:** Update system configurations and financial report templates.

6.2. Limitations & Exclusions

- **LI-1: AI Accuracy:** The AI assistant depends on the clarity of the input voice and background noise. It may require user confirmation for ambiguous commands.
- **LI-2: Hardware:** The system requires an internet connection to process AI commands (Voice-to-Text and RAG).
- **LI-3: Scope of Accounting:** The system supports accounting reports for **household businesses** (Circular 88), not for full corporate tax reporting (Corporate Income Tax for large enterprises).

II. Project Management Plan

1. Overview

1.1. Scope & Estimation (WBS)

The project is divided into 5 main phases following the Scrum framework, executed over 8 weeks (Sprints).

#	WBS Item	Comple xity	Est. Effort (man- days)
1	Initiating & Planning		25
1.1	Analyze Requirements & Circular 88 Logic	High	10
1.2	System Architecture & Database Design	Medium	8
1.3	Setup Infrastructure (Git, Docker, Jira)	Low	7
2	Core Modules Development		85
2.1	Authentication & User Management	Low	10
2.2	Product & Inventory Management	Medium	25
2.3	POS & Order Processing (Mobile)	High	30

2.4	Customer & Debt Management	Medium	20
3	Advanced Features (AI & Accounting)		70
3.1	AI Voice-to-Order Integration	High	30
3.2	Auto-Bookkeeping Logic	High	25
3.3	Compliance Reporting (TT88)	High	15
4	Testing & Stabilization		25
4.1	Integration Testing (Mobile ↔ AI ↔ Backend)	High	15
4.2	User Acceptance Test (UAT)	Medium	10
5	Closing & Deployment		10
5.1	Final Documentation & User Manual	Low	6
5.2	Final Presentation & Deployment	Low	4
	Total Estimated Effort		215

1.2. Project Objectives

- **Timeline:** The project must be finished before Feb 8, 2026 .
- **Quality:**
 - AI voice recognition accuracy > 80% for common sales commands.
 - Accounting reports must strictly follow Circular 88/2021/TT-BTC format.
- **Defect Distributions:**
 - Critical Bugs (System crash): 0 allowed at release.
 - Major Bugs (Feature broken): < 3 allowed at release.

1.3. Project Risks

#	Risk Description	Impact	Probability	Resonse Plans
1	AI Inaccuracy & Ambiguity: The AI assistant may misinterpret local accents, slang, or noisy background audio, leading to incorrect order details (wrong quantity/product).	Users lose trust in the automation and revert to manual notebooks.	High	• Implement a mandatory "Review & Confirm" step (Human-in-the-loop) before finalizing any order. • Design the UI to allow quick manual editing of the draft order.

				• Train the system with specific keywords related to construction materials.
2	Regulatory Non-Compliance: Accounting logic might fail to strictly adhere to Circular 88/2021/TT-BTC regarding form templates or calculation rules, causing legal issues for the owner.	Generated reports become unusable for tax purposes.	Medium	• The BA team must meticulously map requirements directly from the government's official guidelines. • Implement comprehensive Unit Tests for tax and inventory calculation formulas. • Allow export to editable formats (like Excel) for manual correction if needed
3	Low Digital Literacy of Users: Household business owners (Group 1, 2) are often not tech-savvy and may find the mobile interface difficult to navigate.	Slow adoption rate or abandonment of the app.	High	• Prioritize Voice-First interaction (minimizing typing). • Design a minimalist UI with large buttons and high contrast. • Provide simple, video-based onboarding tutorials within the app.
4	Dependency on 3rd Party APIs: The system relies on external services (Google STT, OpenAI) for core features. Latency or API failures can	System hangs or responds slowly during peak times.	Medium	• Implement Optimistic UI (show loading states clearly). • Ensure the "Manual Entry" feature is robust enough to be a

	disrupt the "Voice-to-Order" flow.			standalone backup when AI services are down. • Use caching for frequently ordered products to reduce AI processing tokens.
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2. Management Approach

2.1. Project Process

After evaluating different software development models, the project will adopt the **Scrum Software Development Process**. In this model, the system development is broken down into smaller, manageable cycles called **Sprints**, with each sprint lasting **1 week**. This ensures that functional increments of the software are delivered regularly throughout the 8-week timeline.

- The reasons for choosing this model include:
 - **Developing core features based on priority:** Essential features like POS and Inventory are built first, followed by complex modules like AI and Accounting.
 - **Flexibility to accommodate changes:** Easy to adjust AI prompts or accounting rules (Circular 88) based on testing results.
 - **Continuous testing:** Bugs are identified and fixed within each sprint rather than at the end of the project.
 - **Risk management:** Risks regarding AI accuracy and system integration are identified and addressed early.
 - **Early value delivery:** A working version of the application is available for review at the end of every week.

2.2. Quality Management

2.2.1. Defect Prevention

Each identified defect must be assessed for criticality ("Does it break the POS flow?", "Is the accounting calculation wrong?"). A deadline for resolution must be set. The team adopts a proactive approach by clarifying business rules (Circular 88) before coding to prevent logic errors.

2.2.2. Reviewing

Code reviews are mandatory. Before merging into the `main` branch, Pull Requests (PRs) must be reviewed by at least one other team member. Defects found during review are logged in Jira and must be resolved before merging

2.2.3. Unit Testing

Developers are responsible for writing unit tests for critical functions, especially:

- **Backend:** Tax calculation logic, Inventory deduction logic.
- **AI:** Keyword extraction accuracy (e.g., input "5 bags cement" → output `qty: 5, product: cement`).

2.2.4. Integration Testing

Focuses on the interaction between Mobile App/Web Frontend and the Backend API. Ensures that data sent from the POS screen is correctly stored in the MySQL database.

2.2.5. System Testing

Validate the complete workflow: Voice Input → AI Processing → Draft Order → Confirmation → Accounting Entry → Report Generation.

2.3. Project Deliverables

#	Deliverable	Due Date	Notes
1	Sprint 1	Dec 8, 2025 - Dec 14, 2025	• SRS Document. • Database Design (ERD). • UI/UX Wireframes (Mobile & Website).
2	Sprint 2	Dec 15, 2025 - Dec 21, 2025	• Setup Base Project (Python Clean Arch, Flutter, NextJS). • Authentication module (Login/Register/Auth). • Database implementation.
3	Sprint 3	Dec 22, 2025 - Dec 28, 2025	• Product Catalog management (CRUD). • Multiple Unit Conversion logic (Bao, Tấn, Cái...). • Inventory Stock tracking logic.
4	Sprint 4	Dec 29, 2025 - Jan 4, 2026	• AI Service integration (Voice-to-Text). • Draft Order Logic implemented.
5	Sprint 5	Jan 5, 2026 - Jan 11, 2026	• POS Screen UI (Mobile). • Cart management & Order creation API. • Integration with Bluetooth/LAN printer.
6	Sprint 6	Jan 12, 2026 - Jan 18, 2026	• Customer CRM module. • Debt tracking & Transaction logs implementation. • At-counter order flow completion.
7	Sprint 7	Jan 19, 2026 - Jan 25, 2026	• AI Service Integration - Phase 1: Voice-to-Text (Whisper).

			- Record voice interface on Mobile App. - Real-time notifications (Socket/Push).
8	Sprint 8	Jan 26, 2026 - Feb 1, 2026	AI Service Integration - Phase 2: NLP/LLM for Order Extraction. - Draft Order logic (Review & Confirm flow). - Mapping natural language to Product SKUs.
9	Sprint 9	Feb 2, 2026 - Feb 8, 2026	Compliance Reporting: Export PDF/Excel (S1, S2, S6, S7). - Owner Dashboard (Analytics & Charts). - Admin Portal (Store & Subscription management).
10	Sprint 10	Feb 9, 2026 - Feb 15, 2026	- System Integrated Testing (SIT). - User Acceptance Test (UAT). - Final Documentation, Installation Guide & User Manual.

2.4. Responsibility Assignments

(D~Do; R~Review; S~Support)

Responsibility	Leader	BA / Tester	Backend / AI	Web / Mobile
Project Planning & Tracking	D	S	I	I
Software Requirement (SRS)	R	D	S	S
System Design (DB/API)	S	I	D	S
Coding & Implementation	S	I	D	D
Testing & Bug Reporting	S	D	S	S

User Guides & Final Report	D	D	S	S
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2.5. Project Communications

Communication	Participants	Purpose	Frequency	Tool
Weekly Planning	All Members	Review the previous sprint, schedule, members' work achievements during the week, and report the project progress and status.	Sunday	Discord
Daily Standup	All Members	Quick update on progress and blockers.	Daily	Zalo
Unscheduled Meeting	All Members	When there is a critical problem that needs to be resolved immediately, discuss and then fix that problem.	A critical problem occurred	Discord/Zalo/Offline

2.6. Configuration Management

- **Document Management:** Confluence.
- **Source Code Management:** GitHub.
 - Repository: `software-engineering-project`
 - Branches: `main`, `develop`, `feature/*`

2.7. Tools & Infrastructures

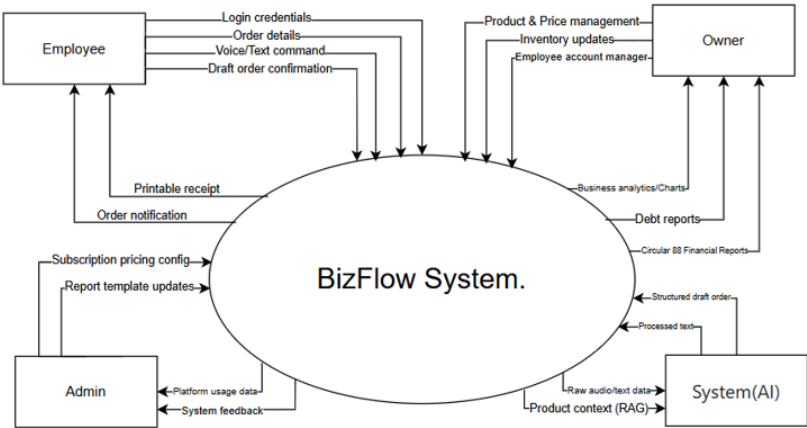
Category	Tools / Infrastructure
Technology	NestJS (Web), Flutter (Mobile), Flask (Backend)
Database	MySQL (XAMPP/Workbench)
IDEs/Editors	Visual Studio Code, Visual Studio
Diagramming	Draw.io, Figma
Project Management	Jira Software

III. Software Requirement Specification

1. Product Overview

BizFlow is a digital platform developed to modernize the operation of household businesses in Vietnam. The system integrates a Point-of-Sale (POS) system with an AI-based assistant to support order processing through voice or text and to assist basic bookkeeping in compliance with Circular 88/2021/TT-BTC.

The platform focuses on traditional household businesses, particularly in construction materials and hardware retail, which are commonly classified under Group 1 or Group 2 according to Decision 3389/QĐ-BTC (2025). BizFlow aims to reduce manual processes, improve operational efficiency, and support regulatory compliance. The system is designed to evolve over future releases with expanded automation and integration capabilities.



2. User Requirements

2.1. Actor

#	Actor	Description
1	Employee	Log into the system, quickly create in-store sales orders by searching products, entering quantities, and adding customer information, print sales receipts, record customer credit for registered clients, receive real-time notifications for new orders, and review and confirm AI-generated draft orders.
2	Owner	Has all employee privileges and additionally manages the product catalog, inventory, and customer data, oversees reports and analytics on sales and outstanding debts, and administers employee accounts.
3	System (AI)	Converts natural language inputs into draft orders and automatically executes accounting operations.

4	Admin	Manages store owner accounts, monitors system reports and analytics, oversees subscription plans and service pricing, and maintains system configurations and financial reporting templates.
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2.2 Use Case

2.2.1. Disagram



2.2.2 Descriptions

ID	Use Case	Actors	Use Case Description
1	Login	Employee	Employee login to the Bizlow system
2	Create order	Employee	Employee creates a new order for customers
3	Print invoice	Employee	Employee prints the invoice from the created order
4	Record debt	Employee	Records debt if the order is not fully paid
5	Confirm draft order	Employee	Confirms draft orders to convert them into official ones
6	Receive order notification	Employee	Employee receives notifications when new orders are placed
7	Manage orders	Owner	The owner monitors and processes orders

8	Manage inventory	Owner	Manages stock-in, stock-out, and product inventory
9	Manage staff	Owner	Manages staff information and permissions
10	Manage customers	Owner	Manages customer profiles and purchase history
11	View revenue report	Owner	Views revenue statistics over time
12	Export tax report	Owner	System exports tax report data for the owner
13	Automatic accounting	AI system	AI system automatically performs accounting entries
14	Convert language to order	AI system	AI analyzes voice/text to generate orders
15	Manage owner accounts	Admin	Admin manages business owner accounts
16	Manage service packages	Admin	Manages software subscription packages
17	View system analytics	Admin	Views overall system activity reports
18	Update tax report templates	Admin	Updates tax forms according to new regulations

3. Functional Requirements

ID	Feature Name	Actor	Detailed Requirements
FE-01	User login	All	Users log into the system using a username/phone number and password. The system verifies credentials and grants access based on roles.
FE-02	Product Search & Filtering	Employee	Employees can quickly search for products by name or category using the search bar or shortcut keys.
FE-03	Create At-Counter	Employee	Employees add items to the cart, adjust quantities, select units, and calculate the

	Order		total amount for walk-in customers.
FE-04	Record Customer Debt	Employee	During checkout, employees can mark the order as "Credit" and assign it to a registered customer.
FE-05	Print Sales Receipt	Employee	The system connects to a printer via Bluetooth/LAN to print sales receipts based on a pre-defined template.
FE-06	Voice-to-Order (AI)	Employee	Users record a voice command. The system converts audio to text and generates a draft order.
FE-07	Review Draft Order	Employee	Users review the AI-generated draft order, edit quantities or products if necessary, and confirm to finalize the order.
FE-08	Manage Product Catalog	Owner	Owners can add, update, delete products, set pricing, and define multiple units of measure.
FE-09	Manage Inventory	Owner	Record stock imports, view real-time stock levels, and track inventory transaction history.
FE-10	Manage Customers	Owner	Add new customers, view purchase history, and track total outstanding debt for each customer.
FE-11	View Business Reports	Owner	View dashboards for daily/monthly revenue, best-selling products, and profit estimates.
FE-12	Compliance Reporting	Owner	Automatically generate and export accounting ledgers (Revenue Ledger, Debt Report) strictly following Circular 88/2021/TT-BTC.
FE-13	Manage Employees	Owner	Create accounts for employees, reset passwords, and deactivate accounts.
FE-14	System Configuration	Admin	Manage subscription plans, update global report templates, and configure system-wide settings.

3.1. Automatic Accounting & Reporting

3.1.1 Description

This function automatically aggregates data from sales and inventory transactions to generate financial reports specifically for household businesses, in accordance with Circular 88/2021/TT-BTC.

3.1.2. Business Rules

ID Rule	Rule Name	Processing Logic
BR-ACC-01	Revenue Recording (Form S1)	Trigger: When Order Status changes to Completed. 1. System extract Total Amount from the order and records it in the Revenue column. 2. Determine the Tax Sector automatically mapping from the product's CategoryID.
BR-ACC-02	Materials & Goods Recording (Form S2)	Trigger: When a Stock Import or Sales Order is approved. Formula: $\text{Closing Stock} = \text{Opening Stock} + \text{Import Quantity} - \text{Export Quantity}$. Quantity is normalized based on <code>conversion_factor</code> defined in <code>product_units</code> table before deduction
BR-ACC-03	Voucher Generation	Logic based on Payment Method: 1. If Cash: Generate Receipt Voucher or Payment Voucher . 2. If Bank Transfer: Record as Bank Credit Note or Debit Note .

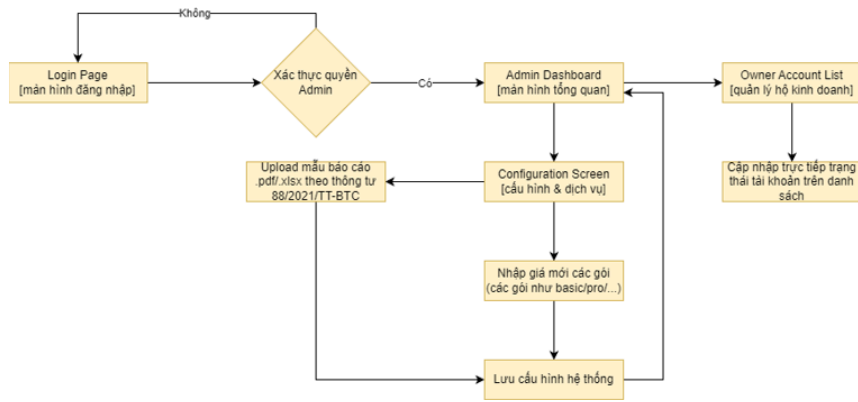
3.1.3. System Functional Overview

- Form S1-HKD: Revenue Ledger.
- Form S2-HKD: Ledger of Materials, Tools, Products, and Goods.
- Form S6-HKD: Cash Book.
- Form S7-HKD: Bank Deposit Book.

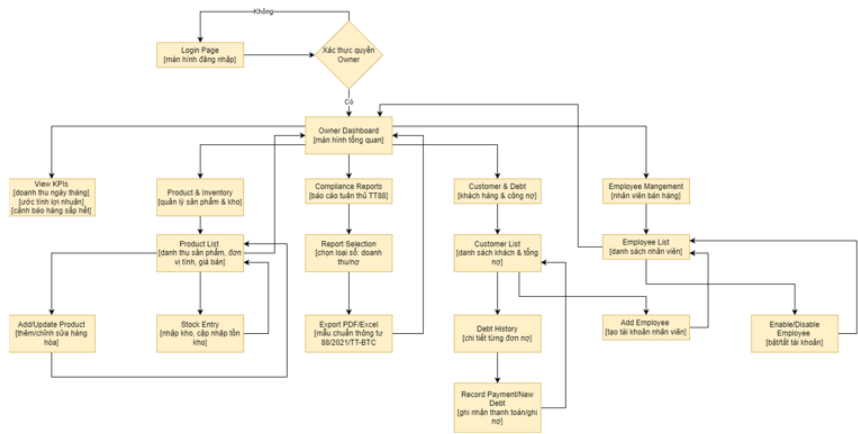
3.2. System Function Overview

3.2.1. Screens Flow

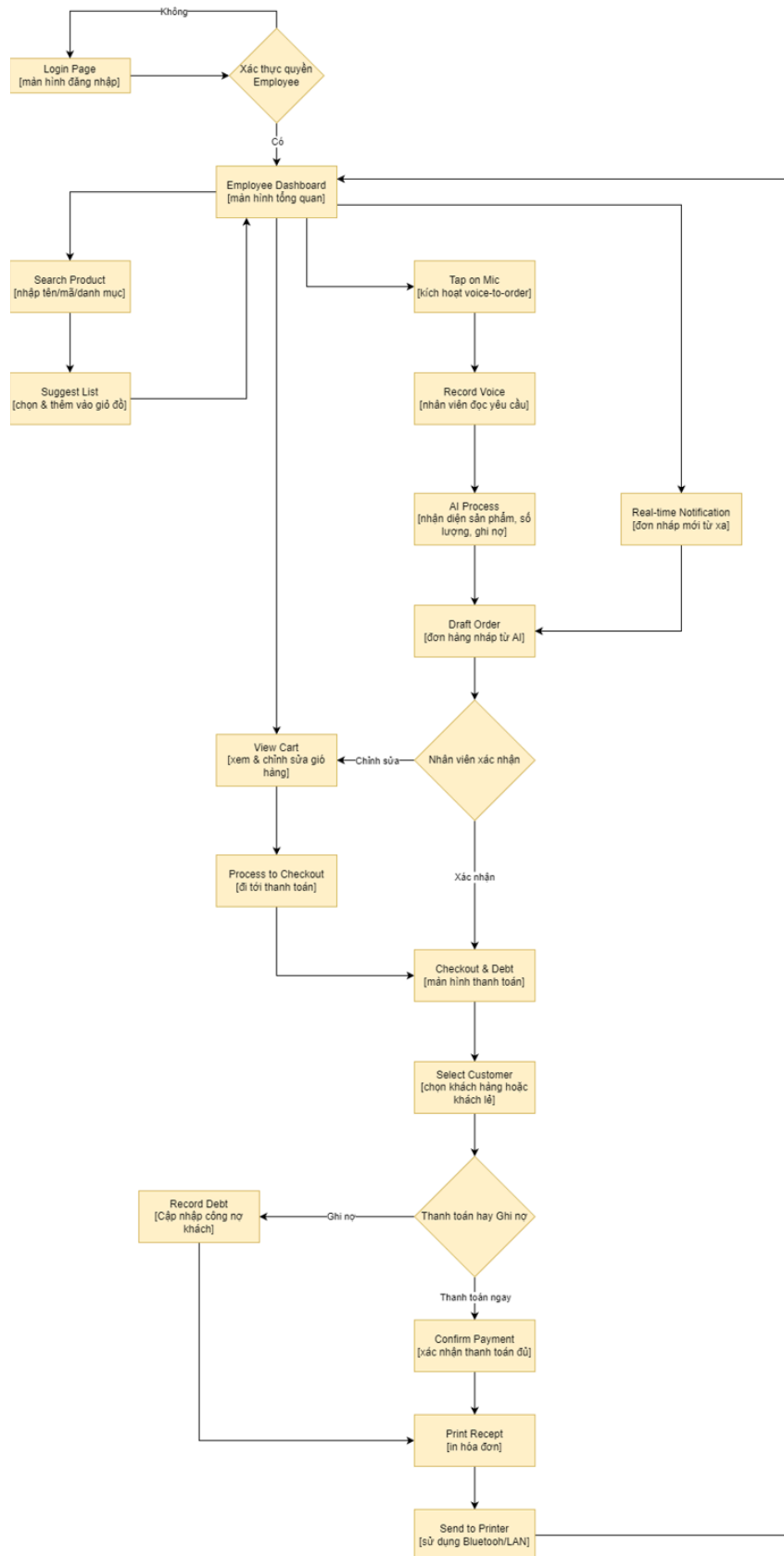
Admin



Owner



Employee



3.2.2. Screen Descriptions

#	Feature	Screen	Description
1	Authentication	Login Page	The entry point for all user. Allows logging in and verifies

			roles (Admin/Owner/Employee) to redirect to the correct dashboad
2	Dashboard	Employee Dashboard	Main workspace for staff. Display shortcuts to Search Product, Voice Order, and recent notifications.
3	Product Search	Search Product	Interface for staff to input product name or scan codes to look up items manually
4	Product Search	Suggest List	Display search results with real-time stock quantities, allowing staff to add item to the cart.
5	Voice Order	Tap on mic	The trigger action to open the voice recording modal/interface.
6	Voice Order	Record Voice	Visualizes audio recording (waveforms) while the employee speaks the order command.
7	AI Processing	AI Process	(System screen) Display a loading state while the system converts voice to text and extracts order details.
8	AI Processing	Draft Order	Display the AI-Generated order (products, quantities) for employee verification before confirming
9	Order Management	View Cart	Show list of selected items. Allows editing quantities or removing items before checkout.
10	Payment	Checkout & debt	The final payment screen showing total amount. Options to proceed with payment or record debt.
11	Payment	Select Customer	A modal or screen to search and select a specific customer for debt recording or loyalty points.
12	Payment	Confirm Payment	Final confirmation step to record the transaction into the system.

13	Output	Print Receipt	Generates a digital receipt view.
14	System	Send to Printer	(Background/Popup) Handles the connection to Bluetooth/LAN printers to print the physical bill.
15	Notification	Real-time Notification	Popup alerts informing staff about new orders or systems updates.
16	Dashboard	Owner Dashboard	Comprehensive view for business owners showing KPIs like Daily Revenue and Profit Estimation.
17	Inventory	Product & Inventory	Menu or section grouping all product - related management features.
18	Inventory	Product List	Detailed table of all products (Unit, Price, Stock). Allows navigation to edit or import stock.
19	Inventory	Add/Update Product	Form to create a new product or update existing details (Price, Name, Unit).
20	Inventory	Stock Entry	Form to record incoming goods from suppliers to update inventory levels.
21	Compliance	Compliance Reports	Menu section for accessing financial reports required by Circular 88.
22	Compliance	Report Selection	Interface to choose the type of report (Revenue Ledger, Inventory Ledger) and date range.
23	Compliance	Export PDF/Excel	Preview and download the selected report in the government -standard format.
24	Customer Relationship Management	Customer & Debt	Menu section for managing customer profiles and outstanding debts.

25	Customer Relationship Management	Customer List	List all customers with their current total debt.
26	Customer Relationship Management	Debt History	Detailed transaction logs for a specific customer (Purchase vs Payment history).
27	Customer Relationship Management	Record Payment/New Debt	Form to record a debt repayment transaction or add a new debt manually.
28	Human Resources	Employee Management	Menu section for managing staff accounts.
29	Human Resources	Employee List	Displays all staff accounts and their status.
30	Human Resources	Add Employee	Form to register a new staff account.
31	Human Resources	Enable/Disabled Employee	Action to toggle access rights for a specific staff account.
32	Dashboard	Admin Dashboard	Global overview for the platform administrator showing total stores and revenue.
33	Tenancy	Owner Account List	Management screen listing all household business accounts registered on the platform.
34	Tenancy	Update Status	Action to activate or deactivate a specific store's subscription.
35	Configuration	Configuration Screen	Settings page for global system parameters and services.
36	Configuration	Update Pricing	Form to modify the subscription prices for Basic/Pro plans.

3.2.3. Screen Authorization

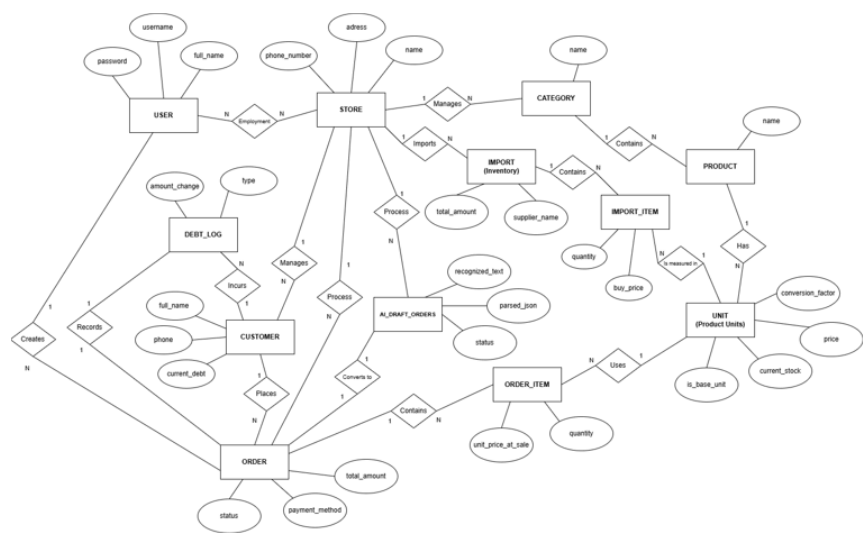
Screen	Admin	Owner	Employee
Login Page	x	x	x
Employee Dashboard		x	x
Search Product		x	x
Suggest List		x	x

Voice Recording (Moda)		x	x
Draft Order (AI results)		x	x
View Cart		x	x
Checkout & Debt		x	x
Select Customer		x	x
Confirm Payment		x	x
Receipt View		x	x
Real-time Notification		x	x
Owner Dashboard		x	
Product & Inventory Menu		x	
Product List		x	x
Add/Update Product		x	
Stock Entry		x	
Compliance Reports Menu		x	
Report Selection		x	
Export PDF/Excel		x	
Customer & Debt Menu		x	
Customer List		x	
Debt History		x	
Record Payment/New Debt		x	
Employee Management Menu		x	
Employee List		x	
Add Employee		x	
Enable/Disable Employee		x	
Admin Dashboard	x		
Owner Account List	x		
Update Status	x		
Configuration Screen	x		
Update Pricing	x		

3.2.4 Non-Screen Functions

#	Feature	System Function	Description
1	Automated Voice Processing	API Service	When the app sends an audio file, the system calls Google Speech-to-Text to transcribe it, then uses an LLM (OpenAI) to extract structured JSON data (Product ID, Quantity, Unit) for the draft order.
2	Automated Stock Deduction	Database Logic/Trigger	When an order is confirmed, the system automatically calculates the conversion factor (e.g., Box to Can) and deducts the precise quantity from the <code>product_units</code> table.
3	Debt Recording	System Logic	If the payment method is "Debt", the system automatically creates a record in <code>debt_transaction_logs</code> and updates the customer's <code>current_debt</code> balance without requiring manual entry.
4	Low Stock Monitoring	Cron Job	A scheduled background task runs daily (e.g., at 08:00 AM) to check stock levels. If a product falls below the threshold, it triggers a "Low Stock" notification to the Owner.
5	Report Generation (TT88)	API Endpoint	The system aggregates data from Orders, Inventory, and Debt logs to generate financial reports (Revenue Ledger S1, Inventory Ledger S2) strictly following the Circular 88/2021/TT-BTC format.

3.3. Entity Relationship Diagram (Business Logic)



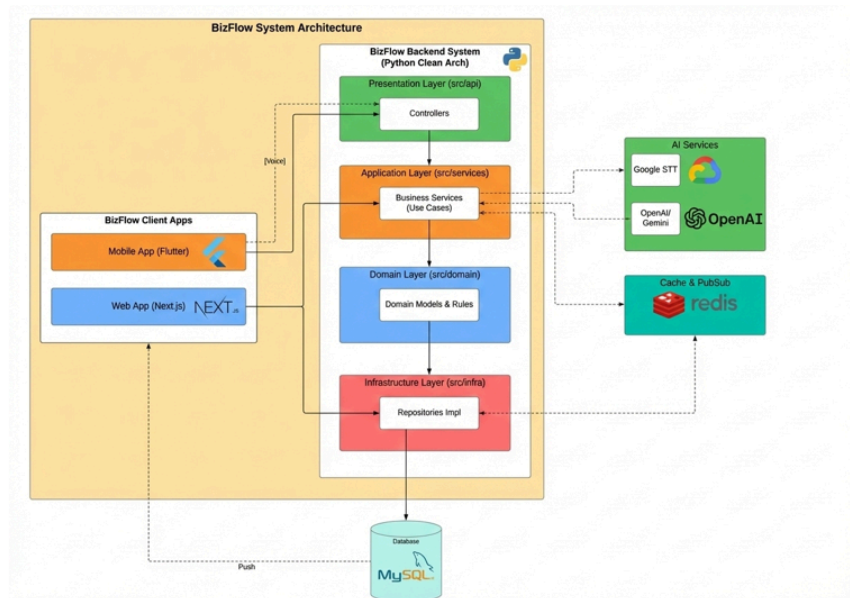
4. Non-Functional

#	Attribute	Specification
NF-01	Usability	The system prioritizes usability and accessibility with a simple, intuitive interface suitable for non-technical users, full Vietnamese language and Unicode support, and real-time notifications.
NF-02	Performance	The system delivers sub-2000 ms response times for core operations and is designed to scale efficiently, supporting large product catalogs and high levels of concurrent users.
NF-03	Reliability & Accuracy	The system ensures AI reliability and accuracy by allowing users to review, edit, or reject AI-generated draft orders, and by supporting full manual operation when the AI is unavailable.
NF-04	Security	The system ensures the security and privacy of household business sales data through strict role-based access control for Employees, Owners, and Administrators.
NF-05	Compliance & Reporting	The system automatically generates accounting reports in compliance with Circular 88/2021/TT-BTC, allows store owners to review, edit, or reject AI-generated reports, and continuously updates report templates to reflect the latest tax regulations.

IV. Software Design Description

1. System Design

1.1. System Architecture



1.1.1. Frontend

- **Next.js (Web Client):** A React framework used for the web dashboard, providing server-side rendering (SSR) for better performance and SEO optimization for the management portal.
- **Flutter (Mobile App):** A cross-platform UI toolkit used to build the mobile application, ensuring a smooth and native-like experience for shop owners on both Android and iOS from a single codebase.
- **TypeScript:** The core programming language for the web frontend, providing static type checking to enhance code quality and maintainability.
- **Shadcn UI & Tailwind CSS:** A modern UI component library and utility-first CSS framework used to design a consistent, responsive, and professional user interface.

1.1.2. Backend

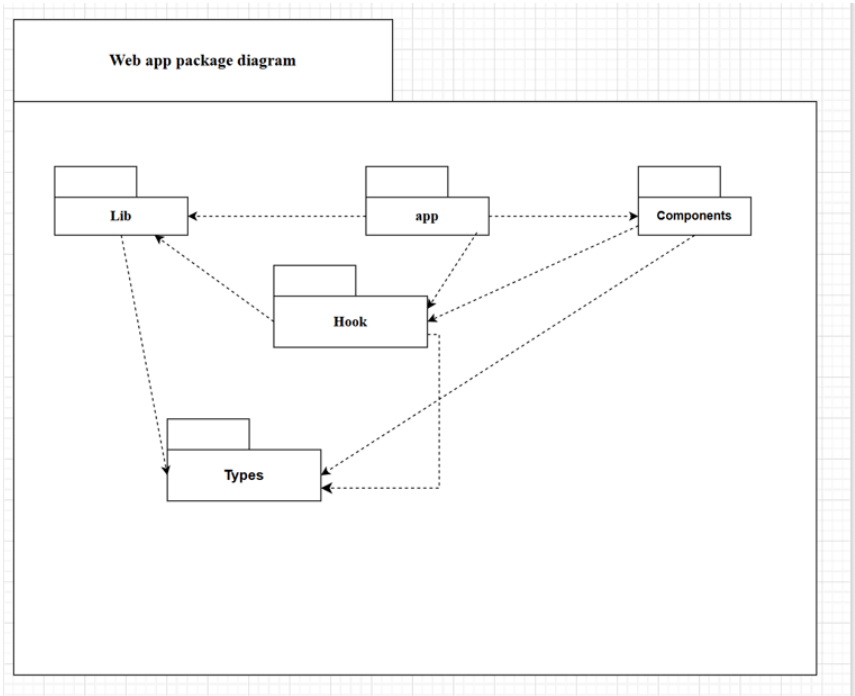
- **Python:** The primary programming language for the backend system, selected for its robust ecosystem supporting AI integration and data processing.
- **Flask:** A modern, high-performance web framework used to build RESTful APIs. It implements **Clean Architecture** to ensure separation of concerns and scalability.
- **MySQL:** Relational database management system (RDBMS), used as the primary storage for structured business data such as users, products, orders, and accounting logs.
- **Redis:** An in-memory data store used for caching frequently accessed data to improve performance and serving as a Message Broker (Pub/Sub) for real-time notifications.
- **ChromaDB:** A vector database used to store data embeddings, supporting Retrieval-Augmented Generation (RAG) to provide context for the AI processing engine.
- **Docker:** A containerization platform used to package the application and its dependencies, ensuring consistent execution environments across development and deployment.

1.1.3. Third-party & AI Services

- **Google Speech-to-Text:** A cloud API service used to transcribe voice recordings from the mobile app into text with high accuracy, specifically optimized for the Vietnamese language.
- **OpenAI API (or Gemini):** The Large Language Model (LLM) service that acts as the system's "brain," analyzing natural language text to extract structured order information (JSON) and automate bookkeeping.
- **Postman:** An API platform used for testing and documenting the backend REST APIs during the development process.

1.2. Package Diagram

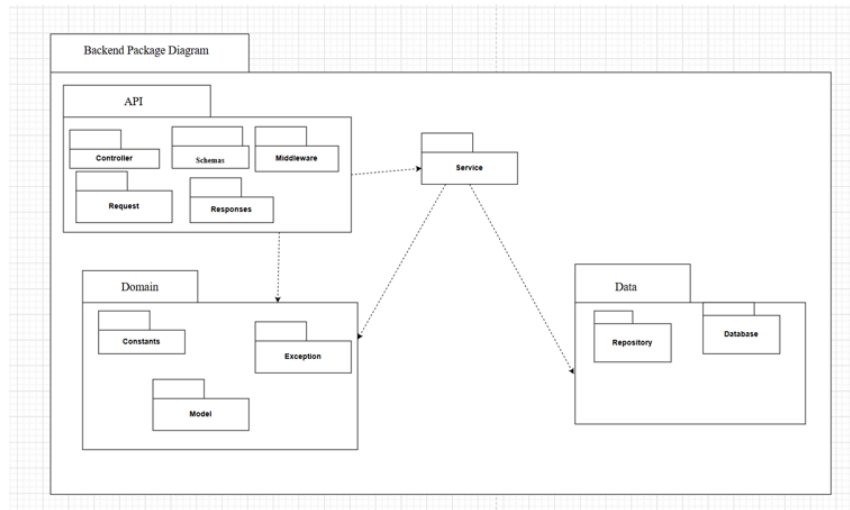
1.2.1. Frontend



No.	Package / Directory	Description
1	src	The root directory containing the entire source code of the frontend application, including modules, UI components, logic, and configurations.
2	src/app	Implements the Next.js App Router structure. This directory manages the application's routing system, global layouts, and page entry points .
3	src/components	Contains all UI components of the application. The components are organized using a Feature-based structure to separate logic by user roles.

4	src/components/ui	Houses atomic, reusable UI elements (e.g., Buttons, Inputs, Modals) built with Shadcn UI to ensure design consistency across the system.
5	src/components/admin	Contains specific dashboards and management interfaces designed exclusively for the System Administrator role.
6	src/components/owner	Contains feature modules for Business Owners, including inventory management, employee oversight, and analytics reports.
7	src/components/employee	Contains operational interfaces for Employees, focusing on order creation and draft management.
8	src/components/common	Stores shared components used across multiple features, such as , ensuring reusability.
9	src/lib	Contains core application logic and utility modules, such as API configuration and authentication handlers .
10	src/data	Stores static used for frontend development and testing purposes before full backend integration.
11	src/types	Defines TypeScript interfaces and type definitions to ensure strict type safety and data consistency throughout the application.
12	src/styles	Contains global stylesheets, including Tailwind CSS directives and global theme variables .

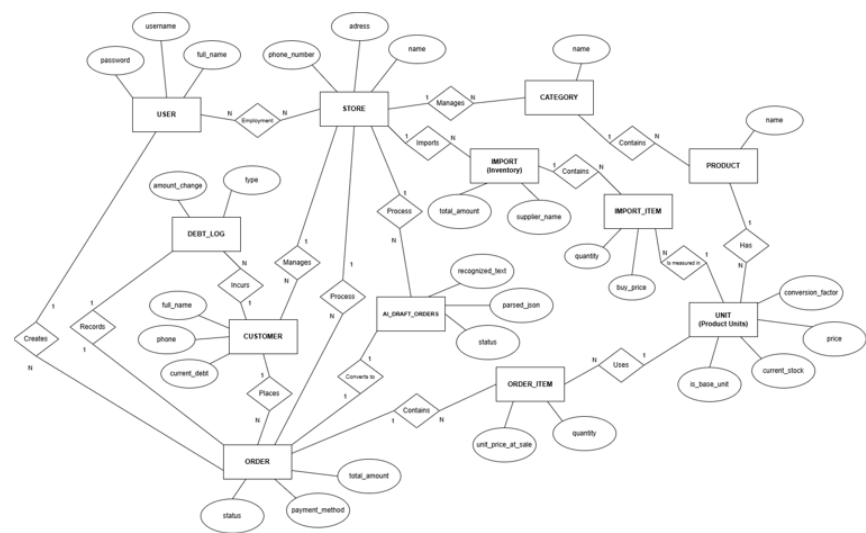
1.2.2. Backend



No.	Package / Directory	Description
1	backend	The root directory containing the entire source code of the backend application, utilizing Flask framework and Clean Architecture.
2	api	Represents the Presentation Layer. Handles HTTP requests, routing, middleware, and input validation.
3	api/controllers	Contains API endpoints logic. Controllers receive requests, validate inputs using schemas, and delegate business logic to the Service Layer.
4	api/schemas	Defines Marshmallow schemas for data serialization and request validation.
5	services	Represents the Application Layer. Contains business logic services that orchestrate operations between the Domain and Infrastructure layers.
6	domain	Represents the Core Layer. It contains enterprise business rules, constants, and exceptions, independent of any external frameworks.
7	domain/models	Contains Business Entities and Repository Interfaces to enforce dependency inversion.
8	infrastructure	Represents the Infrastructure Layer. Implements details such as database

		connections, ORM mapping, and external service integrations.
9	infrastructure/databases	Manages database connections and configurations.
10	infrastructure/models	Defines Database Models (ORM models) that map directly to database tables .
11	infrastructure/repositories	Implements the repository interfaces defined in the Domain Layer. Handles direct data access operations (CRUD) using the ORM.
12	dependency_container.py	Manages Dependency Injection (DI). Registers and injects services and repositories into controllers to ensure loose coupling.
13	create_app.py	Implements the Application Factory pattern to initialize the Flask application, register blueprints, and configure extensions.

2. Database Design



2.1 Table: Users

Field Name	Type	Description	Uniq ue	Not null	PK,F K
Id	bigint	Unique identifier of the user	yes	yes	PK
Username	vachar	Login username (usually phone number)	yes	yes	no
Password_hash	vachar	Encrypted password	no	yes	no
Full_name	vachar	User's full name	no	yes	no
Is_active	tinyint	Account status (1:active, 0:inactive)	no	yes	no

2.2 Table: Stores

Field Name	Type	Description	Unique	Not null	PK,FK
Id	bigint	Unique identifier of the store	yes	yes	PK
Name	vachar	Display name of the store	no	yes	no
Address	text	Business address	no	no	no
Phone	vachar	Hotline number	no	no	no

2.3. Table: Store_user

Field Name	Type	Description	Unique	Not null	PK,FK
User_id	bigint	Reference to user	no	yes	PK,FK
Store_id	bigint	Reference to store	no	yes	PK,FK
Role	enum	User role ('owner', 'staff')	no	yes	no

2.4. Table: Categoryes

Field Name	Type	Description	Unique	Not null	PK,FK
Id	bigint	Unique identifier of the category	yes	yes	PK
Store_id	bigint	Reference to store	no	yes	FK
Name	vachar	Category name	no	yes	no

2.5. Table: Products

Field Name	Type	Description	Unique	Not null	PK,FK
Id	bigint	Unique identifier of the product	yes	yes	PK
Store_id	bigint	Reference to store	no	yes	FK
Category_id	bigint	Reference to category	no	yes	FK
Name	vachar	Product name	no	yes	no
Base_unit	vachar	Base unit name	no	no	no

2.6. Table: Product_units

Field Name	Type	Description	Unique	Not null	PK,FK
Id	bigint	Unique identifier of the unit	yes	yes	PK
Product_id	bigint	Reference to product	no	yes	FK
Unit_name	varchar	Unit name (Bao, Kg, Tan)	no	yes	no
Conversion_factor	decimal(12,3)	Conversion rate relative to base unit	no	yes	no
Price	decimal(15,2)	Selling price for this unit	no	yes	no
Is_base_unit	tinyint	Flag for base unit (1=true)	no	yes	no

2.7. Table: Orders

Field Name	Type	Description	Unique	Not null	PK,FK
Id	bigint	Unique identifier of the order	yes	yes	PK
Store_id	bigint	Reference to store	no	yes	FK
Customer_id	bigint	Reference to customer	no	yes	FK
Creator_user_id	bigint	Staff who created the order	no	yes	FK
Total_amount	decimal(15,2)	Total order amount	no	yes	no
Status	enum	Order status (pending, completed)	no	yes	no

2.8. Table: Order_items

Field Name	Type	Description	Unique	Not null	PK,FK
Id	bigint	Unique identifier of the item	yes	yes	PK

Order_id	bigint	Reference to order	no	yes	FK
Product_unit_id	bigint	Sales unit referece (Bao or Kg)	no	yes	FK
Quantity	decimal (12, 3)	Quantity sold (supports fractions)	no	yes	no
Unit_price_at_sale	decimal (15, 2)	Price at the time of sale	no	yes	no

2.9. Table: Customers

Field Name	Type	Description	Uniq ue	Not null	PK,F K
Id	bigint	Unique identifier of the customer	yes	yes	PK
Store_id	bigint	Reference to store	no	yes	FK
Full_name	vachar	Customer name	no	yes	no
Phone	vachar	Phone number	no	no	no
Current_debt	decimal (15, 2)	Current total debt (Cached field)	no	yes	no

2.10. Table: Debt_transaction_log

Field Name	Type	Description	Uniq ue	Not null	PK,F K
Id	bigint	Unique identifier of the log	yes	yes	PK
Store_id	bigint	Reference to store	no	yes	FK
Customer_id	bigint	Reference to customer	no	yes	FK
Order_id	bigint	Related order ID (if purchasing on credit)	no	no	FK
Amount_change	decimal (15, 2)	Amount changed (positive/negative)	no	yes	no
Transaction_type	enum	Transaction type (increase/decrease)	no	yes	no

2.11. Table: AI_draft_order

Field Name	Type	Description	Unique	Not null	PK,FK
Id	bigint	Unique identifier of the draft	yes	yes	PK
Store_id	bigint	Reference to store	no	yes	FK
Recognized_text	text	Original voice - to - text input	no	yes	no
Parsed_order_json	json	AI - processed result in JSON format	no	yes	no
status	enum	Processing status	no	yes	no

2.12. Table: Inventory_imports

Field Name	Type	Description	Unique	Not null	PK,FK
Id	bigint	Unique identifier of the import slip	yes	yes	PK
Store_id	bigint	Reference to store	no	yes	FK
Supplier_name	vachar	Name of the supplier	no	yes	no
Total_amount	decimal (15, 2)	Total cost of the import batch	no	yes	no
Creaat	datetimee	Import timestamp	no	yes	no

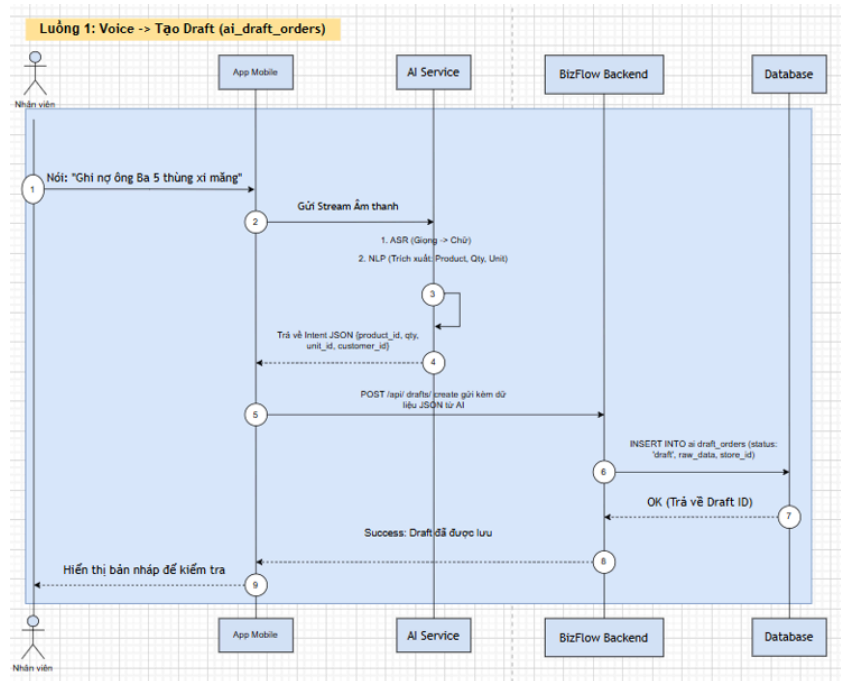
2.13. Table: Inventory_import_items

Field Name	Type	Description	Unique	Not null	PK,FK
Id	bigint	Unique identifier of the import item	yes	yes	PK
Import_id	bigint	Reference to the parent import slip	no	yes	FK
Product_unit_id	bigint	Reference to the specific product unit	no	yes	FK
Quantity	decimal (121. Voice Recogni	Imported quantity	no	yes	no

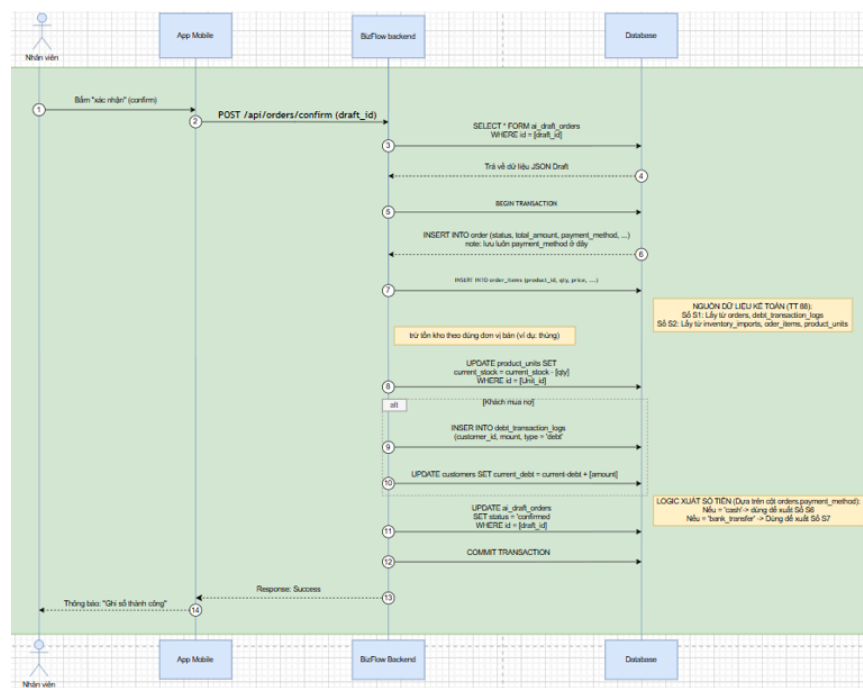
	tion and Draft Order Creatio n Flo, 3)				
Buy_price	decimal (15, 2)	Cost price per unit at the time of import	no	yes	no

3. Detailed Design

3.1. Sequence Diagram for Voice Recognition and Draft Order Creation Flow



3.2 Order Confirmation and Automatic Accounting Flow (Circular 88)



V. Software Testing Documentation

5. Test Reports

5.1 Overview

This chapter presents a summary of the actual testing results conducted on the system after implementation. The testing focuses on validating core functionalities and evaluating user acceptance, especially the effectiveness of the AI-assisted natural language processing feature.

The objective of this chapter is not to enumerate all possible test cases in detail, but to provide representative results that demonstrate system correctness, usability, and practical applicability.

5.2 Test Environment (Summary)

The system was tested in real user environments to closely reflect actual usage conditions:

- Mobile device: iPhone 13
- Desktop device: Dell Laptop (Window)

These environments represent the typical devices used by end users, ensuring realistic evaluation results.

5.3 User Acceptance Testing (UAT) - AI Vietnamese Command Recognition

5.3.1. UAT Description

User Acceptance Testing (UAT) was conducted to evaluate the ability of the AI assistant to understand Vietnamese natural language commands and generate correct draft orders. Users interacted with the application by speaking or entering Vietnamese commands as they would in real business scenarios.

Each AI-generated draft order was manually reviewed by users before confirmation, following the human-in-the-loop design principle.

5.3.2 Test Scenario

- Number of test commands: 10 Vietnamese commands
- Input type: Voice/Text
- Evaluation criteria:
 - Correct identification of product name
 - Correct quantity extraction
 - Correct recognition of customer and payment type (cash/debt)

5.3.3 UAT Result

No.	Vietnamese Command	Expected Interpretation	AI Result	Evaluation
1	Lấy cho anh Ba 5 bao xi măng, ghi nợ	Xi măng - 5 bao- ghi nợ	Parsed correctly	Correct
2	Bán 2 thùng sơn Dulux cho chị Lan	Sơn Dulux- 2 thùng	Parsed correctly	Correct
3	Cho tôi 10 cây thép phi 16	Thép phi 16- 10 cây	Missing unit detail	Incorrect
4	Lấy 3 bao cát cho chú Tư, trả tiền mặt	Cát- 3 bao- tiền mặt	Parsed correctly	Correct
5	Bán anh Nam 1 máy khoan, ghi sổ	Máy khoan- 1-ghi nợ	Parsed correctly	Correct
6	Lấy 20 viên gạch đỏ	Gạch đỏ - 20 viên	Missing customer	Incorrect
7	Bán 5 bao xi măng Holcim	Xi măng Holcim- 5 bao	Parsed Correctly	Correct
8	Cho chị Hạnh 2 thùng sơn, để cuối tháng trả	Sơn- 2 thùng - ghi nợ	Parsed correct	Correct
9	Lấy 1 cuộn dây điện	Dây điện Cadivi - 1	Parsed correctly	Correct

	Cadivi	cuộn		
10	Bán cho anh Ba ít xi	Xi măng- số lượng không rõ	Quantity ambiguous	Incorrect

5.3.4 UAT Summary

- Total test commands: 10
- Commands correctly understood by AI: 7
- Commands incorrectly or partially understood: 3
- AI recognition accuracy: 70%

5.3.5 Observations

- The AI assistant performs effectively when commands contain clear quantities, product names, and customers.
- Most errors occur when commands are ambiguous or lack sufficient detail.
- The human confirmation step successfully prevents incorrect draft orders from affecting official data.

5.4. Representative Functional Test Result

To keep the testing documentation concise, only representative test cases for core features are summarized below.

TC ID	Feature	Test Description	Result
TC-01	Authentication	Login with valid account	Pass
TC-02	Manual Order	Create sales order manually	Pass
TC-03	AI Draft Order	Create order via AI command	Pass
TC-04	Confirm Draft	Confirm AI-generated order	Pass
TC-05	Inventory	Inventory updated after sale	Pass
TC-06	Debt Management	Record customer debt	Pass
TC-07	Payment	Record debt payment	Pass
TC-08	Customer	Add new customer	Pass
TC-09	Product	Disable product	Pass
TC-10	Reporting	Generate daily revenue report	Pass

TC-11	Authorization	Role-based access control	Pass
TC-12	Notification	Recei AI draft notification	Pass
TC-13	Audit Log	Log order creation	Pass
TC-14	Stability	Concurrent order processing	Pass
TC-15	Fallback	Manual operation when AI offline	Pass

5.5 Conclusion

The testing results demonstrate that the sytem satisfies core functional requirements and achieves acceptable AI accuracy for real-world usage. Although the AI assistant does not fully understand all ambiguous Vietnamese commands, the implemmmented human-in-the-loop mechanism ensures operational safety and reliability. Overall, the system is ready for deployment and further enhancement.

VI. Release Package & User Guides

This chapter provides installation and deployment instructions for the BizFlow system, including the web application (Next.js) and Mobile Application (Flutter). The guidelines are written for development, testing, and end-user usage scenarios, consistent with the structure of the submitted Capstone Project document.

6.1. Deliverable Package

The deliverable package submitted for this project includes the following items:

- Source Code
 - Backend source code
 - Web application source code (Next.js)
 - Mobile application source code (Flutter)
- Project Documentation
 - Software Requirement Specification (SRS)
 - Software Design Description (SDD)
 - Software Testing Documentation
 - Installation Guides & User Manual
- Presentation Materials
 - Final presentation slides

6.2. Web Application Installation Guide (Next.js)

6.2.1 System Requirements

- Operating System: Windows/macOS/Linux
- Node.js: version18 or higher

- Package Manager: npm or yarn
- Web Browser: Google Chrome (recommended)

6.2.2 Installation Steps

- Step 1: Clone the source code repository

Download the web application source code from the project repository.

- Step 2: Install Dependencies

Open a terminal in the web project directory and run:

```
1 npm install
```

or

```
1 yarn install
```

- Step 3: Configure environment variables

create a .env.local file and configure required environment variables such as:

API base URL

Authentication keys

- Step 4: Run the application in development mode

Execute the following command:

```
1 npm run dev
```

or

```
1 yarn dev
```

The application will be available at: <http://localhost:3000>

6.2.3. Build for Production

To generate a production build, run:

```
1 npm run build
```

or

```
1 yarn build
```

After building, the application can be deployed to platform such as Vercel or other hosting services.

6.3. Mobile Application Installation Guide (flutter)

6.3.1. System Requirements

- Operating System: Windows/macOS
- Flutter SDK: version 3.x or higher
- Android SDK and Android Studio
- Device: Android phone or emulator

6.3.2. Installation Steps

- Step 1: Install Flutter SDK

Download and install Flutter from [the official website](#). After installation, verify setup using:

```
1 flutter doctor
```

- Step 2: Get project dependencies

Navigate to the mobile project directory and run:

```
1 flutter pub get
```

- Step 3: Run the application in debug mode

Connect an Android device or start an emulator, then run:

```
1 flutter run
```

6.4 Build Android APK for End Users

This section describes how to generate an Android .apk file so that household business owners can install the application directly on their devices.

- Step 1: Configure application information

Update the application name, icon, and version in pubspec.yaml and AndroidManifest.xml as needed.

- Step 2: Build APK file

Run the following command in the Flutter project directory:

```
1 flutter build apk --release
```

- Step 3: Locate APK output

After the build process completes successfully, the APK file will be generated at:

```
build/app/outputs/flutter-apk/app-release.apk
```

- Step 4: Install APK on Android device

Copy the APK file to the Android device and enable "Install unknown apps" in device settings. The application can then be installed and used by the household business owner.

6.5 Notes

- The mobile application is designed for simple usage with minimal configuration to support users with low digital literacy.
- For security reasons, production deployment should include proper signing and version control.
- Further updates can be distributed by providing newer APK versions to users.